

CHAPTER THREE

MATERIALS AND METHODS

3.2 Location of the Study

The experiment will be conducted at the University of Abuja Teaching and Research Farm, located along the Abuja–Lokoja Expressway, Airport Road, Gwagwalada Area Council, Federal Capital Territory (FCT), Nigeria. The farm serves as a major center for agricultural research and practical training. The study area lies within the Guinea Savannah agro-ecological zone of Nigeria.

3.3 Climate of the Study Area

Abuja experiences a tropical wet-and-dry climate characteristic of the Guinea Savannah zone. The area has two major seasons: the rainy season (April–October) with peak rainfall between July and September, and the dry season (November–March). Annual rainfall ranges between 900 and 1500 mm, while mean daily temperatures range from 25°C to 32°C. Relative humidity is high during the rainy season and low during the Harmattan period. These climatic conditions are favorable for watermelon production.

3.4 Vegetation and Human Activities

The natural vegetation is Guinea Savannah, characterized by tall grasses with scattered trees such as *Parkia biglobosa* and *Vitellaria paradoxa*. Human activities include crop farming, livestock grazing, irrigation farming, and agricultural research, which support soil management practices and agronomic experiments.

3.5 Geology and Soil of the Study Area

The geology of Abuja is dominated by Precambrian Basement Complex rocks (Falconer, 1911; Durotoye, 1983). The soils are mainly sandy-loam to loamy, well-drained, slightly acidic to neutral, and low to moderate in fertility. These soils respond positively to soil amendments such as biochar and inorganic fertilizers.

3.6 Preparation of Maize Cob Biochar

Maize cobs were collected from the University of Abuja Teaching and Research Farm, cleaned to remove impurities, and sun-dried. Biochar was produced through slow pyrolysis using a metal drum under limited oxygen supply at 350–500°C. The biochar was cooled, crushed, sieved, and stored in airtight containers prior to application.

3.7 Fertilizer Application

Biochar was applied two weeks before planting and incorporated into the top 10–15 cm of soil. NPK (20:10:10) fertilizer was applied using the ring method in two split doses at 2 and 5 weeks after planting (WAP), lightly covered with soil to prevent nutrient loss.

3.9 Treatment Combination and Experimental Layout

The experiment consisted of a factorial combination of maize cob biochar, NPK (20:10:10) fertilizer, and weeding intervals, arranged in a Randomized Complete Block Design (RCBD) with three replications.

3.9.1 Treatment Factors

Factor A: Biochar application (B₀: 0 t ha⁻¹, B₁: Biochar applied)

Factor B: NPK fertilizer (F₀: 0 kg ha⁻¹, F₁: Applied)

Factor C: Weeding interval (W₁: 2-week, W₂: 4-week).

3.9.2 Experimental Layout

Each block contained all treatment combinations, totaling 24 plots (8 treatments × 3 replications). Buffer zones were provided to minimize interference.

3.10 Land Preparation

The field was manually cleared and tilled to a depth of 20–25 cm. Ridges were constructed at 1 m spacing. Treatments were incorporated into the soil before planting.

3.11 Harvesting

Harvesting was carried out at physiological maturity (10–12 WAP), indicated by drying tendrils, dull rind color, and hollow sound when tapped. Fruits were harvested manually and sorted into marketable and unmarketable categories.

3.12 Data Collection

Data were collected on growth, yield, and soil properties.

3.12.1 Growth Parameters

Plant establishment, vine length, number of leaves, number of vines, and days to flowering were recorded at specified intervals.

3.12.2 Yield Parameters

Number of fruits per plant, fruit weight, fruit length, fruit diameter, total yield, and marketable yield were measured at harvest.

3.12.3 Soil Sampling and Analysis

Soil samples were collected before planting and after harvest using a soil auger at 0–30 cm depth. Samples were bulked into composite samples and analyzed for particle size distribution and pH using standard laboratory methods.